

VOLODYMYR-VOLYNSKY, Ukraine

WMO: 331770

Lat: 50.8331N

Lon: 24.3436E

Elev: 194

StdP: 99.02

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-14.5	-20.7	0.6	-18.1	-17.1	0.9	-14.3	12.3	1.4	9.9	-0.2	1.1	120	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.0	20.5	28.3	19.8	26.6	19.0	21.6	27.9	20.7	26.5	19.8	25.2	2.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.5	24.8	18.5	13.7	23.5	17.7	13.0	22.5	63.8	28.2	60.4	26.6	57.3	25.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.9	5.4	-21.6	32.8	4.7	1.5	-25.0	33.8	-27.7	34.7	-30.3	35.5	-33.7	36.6		
			-21.8	23.3	4.6	0.9	-25.1	23.9	-27.9	24.4	-30.5	24.9	-33.8	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-2.9	-1.6	2.7	9.2	14.2	17.4	19.4	18.6	13.7	8.6	3.7	-1.1
	DBStd	9.00	5.76	5.84	4.75	4.16	3.74	3.22	3.00	3.07	3.59	4.18	4.67	5.30
	HDD10.0	1651	401	325	229	63	8	0	0	0	8	78	194	345
	HDD18.3	3718	659	558	486	274	137	57	24	34	143	302	439	603
	CDD10.0	1114	0	0	2	39	137	222	291	266	120	34	4	0
	CDD18.3	140	0	0	0	1	8	28	57	42	6	0	0	0
	CDH23.3	1401	0	0	0	12	107	267	529	421	64	1	0	0
	CDH26.7	353	0	0	0	1	15	52	149	126	9	0	0	0
Wind	WSAvg	2.2	2.5	2.5	2.7	2.4	2.1	1.9	1.7	1.7	1.8	2.1	2.4	2.4
Precipitation	PrecAvg	647	38	36	38	45	69	79	95	65	60	45	40	37
	PrecMax	778	84	70	82	87	136	169	205	224	132	127	79	69
	PrecMin	504	5	7	8	15	21	21	14	2	9	6	2	16
	PrecStd	76	20	14	19	19	29	40	51	44	36	31	19	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	11.5	17.1	24.8	28.6	30.2	32.0	32.7	27.8	22.2	16.4	9.7
		MCWB	6.4	7.9	10.4	15.3	18.4	20.7	21.5	20.9	18.7	16.2	12.0	7.8
	2%	DB	6.6	8.5	13.7	21.6	26.2	28.3	30.2	29.9	25.2	19.5	13.6	7.7
		MCWB	4.9	6.2	8.4	13.3	17.5	19.8	20.8	20.3	17.7	14.5	10.8	6.2
	5%	DB	5.0	6.6	11.4	18.8	24.1	26.4	28.3	27.7	22.8	17.2	11.6	6.1
		MCWB	3.6	4.8	7.3	12.0	16.7	18.9	20.2	19.7	16.6	13.5	9.7	4.9
	10%	DB	3.6	5.0	9.2	16.6	21.9	24.6	26.4	25.5	20.5	15.0	9.9	4.8
		MCWB	2.5	3.7	6.4	10.8	15.4	18.1	19.4	18.6	15.5	12.0	8.4	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	8.2	10.9	16.1	19.6	22.0	23.1	22.7	20.0	16.7	12.4	8.1
		MCDB	8.1	10.6	16.3	23.8	26.1	27.9	29.6	29.6	26.6	20.7	15.1	9.4
	2%	WB	5.0	6.4	9.1	14.2	18.2	20.7	21.9	21.3	18.4	15.2	11.2	6.3
		MCDB	6.4	8.3	12.8	20.0	24.4	26.6	28.2	27.7	23.5	18.7	13.2	7.5
	5%	WB	3.7	5.0	7.8	12.8	17.2	19.6	20.9	20.2	17.1	13.8	9.8	5.0
		MCDB	4.8	6.3	10.8	17.5	23.2	25.0	26.7	26.0	21.4	16.7	11.6	6.0
	10%	WB	2.6	3.6	6.5	11.5	16.1	18.6	20.0	19.3	16.0	12.3	8.4	3.8
		MCDB	3.5	5.0	9.0	15.5	20.7	23.3	25.2	24.4	19.9	14.7	9.8	4.7
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.1	9.7	10.5	10.2	10.4	11.0	9.5	7.9	5.1	4.4
		MCDBR	4.8	6.7	11.3	13.7	14.6	13.5	14.1	14.9	14.0	11.5	8.6	5.2
		MCWBR	4.0	4.9	6.9	6.9	7.1	6.5	6.2	6.6	7.3	7.0	5.9	4.2
	5% WB	MCDBR	4.4	6.4	10.0	12.2	12.9	12.1	12.3	12.5	12.3	10.8	7.4	5.0
		MCWBR	3.8	5.0	6.6	6.5	6.7	6.4	6.0	6.1	7.0	7.0	5.3	4.3
Clear-Sky Solar Irradiance	taub	0.313	0.342	0.380	0.431	0.419	0.415	0.443	0.441	0.404	0.366	0.349	0.310	
	taud	2.221	2.182	2.190	2.126	2.223	2.260	2.198	2.199	2.259	2.340	2.299	2.274	
	Ebn at Noon	688	766	807	805	837	843	810	789	776	738	642	634	
	Edh at Noon	72	98	117	140	133	130	136	129	109	84	68	60	
All-Sky Solar Radiation	RadAvg	0.83	1.63	2.80	4.18	5.24	5.64	5.36	4.82	3.28	1.95	0.86	0.58	
	RadStd	0.14	0.24	0.37	0.43	0.51	0.41	0.46	0.34	0.48	0.32	0.09	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	+0.57	N/A	N/A	+0.90	+0.48	+0.42	N/A	-187	+102	+42

Nomenclature: See separate page